

Problem

The coefficient of coupling for two coils having $L_1 = 2 \text{ H}$, $L_2 = 8 \text{ H}$, $M = 3 \text{ H}$ is:

- (a) 0.1875
- (b) 0.75
- (c) 1.333
- (d) 5.333

Step-by-step solution

Step 1 of 2

Calculate the coefficient of coupling k for two coils.

$$k = \frac{M}{\sqrt{L_1 L_2}} \dots\dots (1)$$

Here, L_1 and L_2 are the inductances of two coils and M is the mutual inductance between these coils.

Step 2 of 2

Recall equation (1).

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

Substitute 3 for M , 2 for L_1 , and 8 for L_2 .

$$\begin{aligned} k &= \frac{3}{\sqrt{(2)(8)}} \\ &= \frac{3}{4} \\ &= 0.75 \end{aligned}$$

Therefore, the coefficient of coupling k for two coils is **0.75**.

Hence, the correct option is **(b)**.